

Siddhartha Phuyal

LinkedIn: [linkedin.com/in/siddhartha-phuyal-386662132](https://www.linkedin.com/in/siddhartha-phuyal-386662132)

Github: github.com/siddhartha201

Email : phuyalsiddhartha@gmail.com

Mobile : +1-6055212376

EDUCATION

- **University of South Dakota** Vermillion, SD
Master of Science in Computer Science; GPA: 3.9 *Jan. 2024 – Dec. 2025*
- **The British College** Kathmandu, Nepal
Master of Business Administration *Sept. 2022 – Aug. 2024*
- **KIIT University** Bhubaneswar, India
Bachelor of Technology in Civil Engineering; GPA: 7.44/10.0 *Aug. 2016 – Jul. 2020*

WORK EXPERIENCE

- **University of South Dakota** Vermillion, SD
Research Assistant *Jan. 2024 – Dec. 2025*
 - **Deep Learning Research:** Developed and evaluated deep learning models using TensorFlow and PyTorch across six academic and research projects in computer vision, healthcare AI, and medical image analysis.
 - **Model Development:** Implemented CNNs, transfer learning architectures, multimodal fusion frameworks, and machine learning pipelines using TensorFlow, Keras, and Scikit-learn.
 - **Natural Language Processing:** Tokenized and processed text data for language modeling tasks and implemented Skip-Gram, CBOW, Seq2Seq, RNN-based, and Transformer-based architectures using PyTorch.
- **Green Apple Incorporation** Kathmandu, Nepal
Educational Consultant *Dec. 2022 – Jul. 2023*
 - **Student Counseling:** Counseled hundreds of students on overseas education opportunities and successfully facilitated visa approvals for 10 students.
- **Golden Goods Construction Private Ltd.** Dorpa, Khotang, Nepal
Site Engineer *Dec. 2020 – May 2022*
 - **Project Management:** Directed engineering activities at Mid-hill Highway project sites, including scheduling daily work for a 20+ member crew, coordinating with contractors, and monitoring progress to ensure timely completion of 40 km of roadway.
 - **Engineering Solutions:** Designed and supervised slope stabilization measures using gabion retaining walls and improved drainage systems to address frequent landslides and water-logging issues along the highway at ten critical locations.
 - **Testing and Observation:** Conducted soil classification and compaction tests, observed concrete curing, and verified asphalt layer thickness to ensure compliance with Nepal Road Standards.

PROJECTS

- **Binary Image Classification:** Built a binary image classification model using TensorFlow's Keras API, tailored for medical imaging (Chest X-rays), achieving an accuracy of 77.8% (21/27 correct predictions).
- **Feature Extraction and Fine-Tuning:** Performed feature extraction and fine-tuning using VGG19 and ResNet50 on skin cancer image data, improving classification performance over baseline CNN models.
- **Diabetes Risk Prediction:** Analyzed and evaluated machine learning models (Decision Trees, Random Forest, XGBoost, LightGBM, CatBoost) for predicting diabetes risk, with LightGBM achieving the best performance (F1 score = 0.805 and 97% cross-validation accuracy).
- **Multi-modal Model Fusion:** Designed a multi-modal model fusion framework combining Ultrasound images and textual patient data for breast lesion analysis; late fusion emerged as the most robust and clinically applicable approach, achieving 98% accuracy and an AUC of 1.00.
- **Vehicle Number Plate Detection:** Implemented vehicle number plate detection using YOLO v8 for object detection and OCR (Optical Character Recognition) for state identification, accurately localizing plates and recognizing state-level classes across 51 categories.
- **Handwritten Digit Recognition:** Implemented a Convolutional Neural Network (CNN) on the MNIST dataset to classify hand-written digits, achieving 98% test accuracy.

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, JavaScript, C
- **Machine Learning & AI:** TensorFlow, Keras, PyTorch, Scikit-learn, Deep Learning, Machine Learning, Computer Vision, Natural Language Processing, Large Language Models (LLMs)
- **Data Science:** Pandas, NumPy, Matplotlib, Data Mining, Data Analysis, Feature Engineering
- **Computer Vision:** OpenCV, YOLOv8, OCR, Image Classification, Object Detection
- **Software & Tools:** Git, GitHub, AutoCAD, Maket (AI-powered Architectural Design Platform)
- **Civil Engineering:** Road Construction, Quantity Estimation, Soil Testing, Project Scheduling, Cost Estimation, Construction Management, Surveying

CERTIFICATIONS

- **Supervised Machine Learning: Regression and Classification:** An online non-credit course authorized by DeepLearning.AI and Stanford University, offered through Coursera.
- **Advanced Learning Algorithms:** An online non-credit course authorized by DeepLearning.AI and Stanford University, offered through Coursera.
- **Macroeconomics:** Udemey certificate in “The Macroeconomics.”
- **Financial Modeling and Valuation in Microsoft Excel:** Udemey certificate in “Financial Modeling and Valuation in Microsoft Excel.”
- **Civil Engineering License:** Licensed Civil Engineer from the Nepal Engineering Council.
- **Generative AI Engineering with Large Language Model Specialization:** An online course authorized by IBM and offered through Coursera.
- **Basic Design of Reinforced Concrete Structures:** An online course by Er. Hardik Solanki offered through Udemey.
- **Vedic Astrology:** Vedic Astrology certificate from Magha Vedic Astrology Academy.

LANGUAGES

- **Nepali:** Native proficiency.
- **English:** IELTS: Reading 7.5, Writing 7, Speaking 7, Listening 6, Overall 7.
- **Hindi:** Fluent in both understanding and speaking.

PARTICIPATION AND EXTRACURRICULAR ACTIVITIES

- **AI Symposium:** Participated in the AI Symposium organized by the Department of Computer Science at the University of South Dakota. Attended presentations by industry experts, researchers, and academicians on recent developments in Artificial Intelligence and its applications across various disciplines for advanced research and innovation.
- **Soccer:** Represented the University of South Dakota in a soccer match against South Dakota State University. An active soccer player with experience participating in numerous competitive tournaments, demonstrating teamwork, leadership, discipline, and collaboration skills.